HUMAN DEVELOPMENT INDEX (HDI) PREDICTION USING MACHINE LEARNING

Project Report

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We sincerely express our gratitude to our internship organization for providing us with the opportunity to work on the Human Development Index (HDI) Prediction project. This project helped us gain practical knowledge in Machine Learning, Data Preprocessing, Model Building, and Flask Web Development.

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Finally, we thank everyone who directly or indirectly contributed to the successful completion of this project.

ABSTRACT

The Human Development Index (HDI) is a statistical measure developed by the United Nations Development Programme (UNDP) to evaluate the overall development of countries. It combines three major dimensions: Life Expectancy, Education, and Gross National Income (GNI) per capita.

This project develops a Machine Learning model capable of predicting the HDI score using these four important indicators:

- Life Expectancy
- Expected Years of Schooling
- Mean Years of Schooling
- Gross National Income Per Capita

The project uses Random Forest Regression for prediction and Flask Framework to develop a user-friendly web application.

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1. INTRODUCTION

The Human Development Index (HDI) is one of the most widely used indicators for measuring the development level of countries. It focuses on improving people's quality of life rather than considering economic growth alone.

The HDI combines health, education, and standard of living into a single numerical value ranging from 0 to 1. Countries with higher HDI scores generally have better healthcare systems, higher educational attainment, and stronger economies.

This project predicts HDI using Machine Learning techniques and provides an easy-to-use web interface for users.

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2. PROBLEM STATEMENT

Predicting Human Development Index manually requires analyzing multiple development indicators and large datasets.

The objective of this project is to automate HDI prediction using Machine Learning based on four important development indicators.

3.OBJECTIVES

The objectives of this project are:

- Predict HDI Score accurately.
- Reduce manual calculations.
- Build a Machine Learning prediction model.
- Develop a Flask web application.
- Improve prediction efficiency.

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4. EXISTING SYSTEM

Traditional HDI analysis is performed using statistical reports and manual calculations.
Limitations include:

- Time consuming
- Large datasets
- Manual analysis
- Difficult for ordinary users

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5. PROPOSED SYSTEM

The proposed system uses Machine Learning to predict HDI score instantly.
The user enters:

- Life Expectancy
- Expected Years of Schooling
- Mean Years of Schooling
- GNI per Capita

The trained Random Forest Regression model predicts the HDI score.

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6. DATASET DESCRIPTION

Dataset Source:
United Nations Development Programme (UNDP)
Dataset Format:
CSV
Target Variable:
Human Development Index (2021)
Input Features:

- Life Expectancy at Birth (2021)
- Expected Years of Schooling (2021)
- Mean Years of Schooling (2021)
- Gross National Income Per Capita (2021)

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7. SYSTEM REQUIREMENTS

Hardware

Processor:
Intel Core i3 or above
RAM:
4 GB minimum
Storage:
10 GB

Software

Python
VS Code
Flask
Scikit-Learn
Pandas
NumPy
Matplotlib

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8. METHODOLOGY

The project follows these steps:

1. Dataset Collection
2. Data Cleaning
3. Feature Selection
4. Data Preprocessing
5. Train-Test Split
6. Random Forest Training
7. Model Saving
8. Flask Integration
9. Prediction

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9. DATA PREPROCESSING

The dataset contains missing values and unnecessary columns.
The preprocessing steps include:

- Removing unwanted columns
- Handling missing values
- Selecting required features
- Splitting training and testing data

10. MACHINE LEARNING MODEL

Algorithm Used:
Random Forest Regression
Reason for Selection:

- High Accuracy
- Handles Non-linear Data
- Robust Performance
- Less Overfitting

Model Accuracy:
99.5%

11.SYSTEM ARCHITECTURE

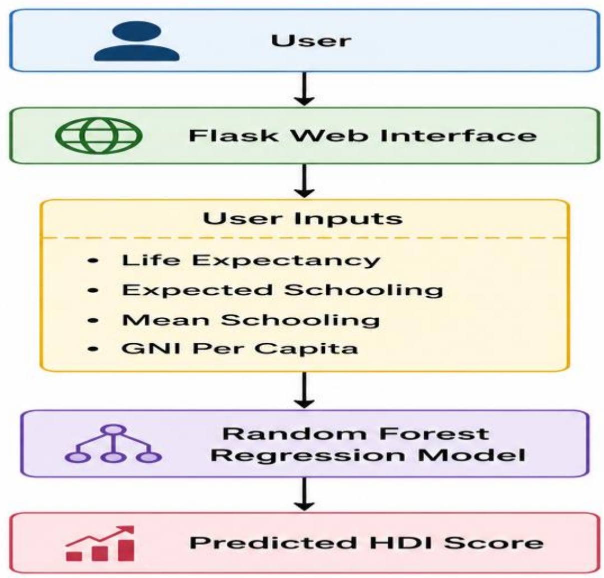

Figure 1.1: System Architecture of HDI Prediction System

Description: The system starts with user input through the Flask web application. The input values are passed to the trained Machine Learning model, which predicts the HDI score. Finally, the prediction result is displayed on the web page.

12. ENTITY RELATIONSHIP DIAGRAM

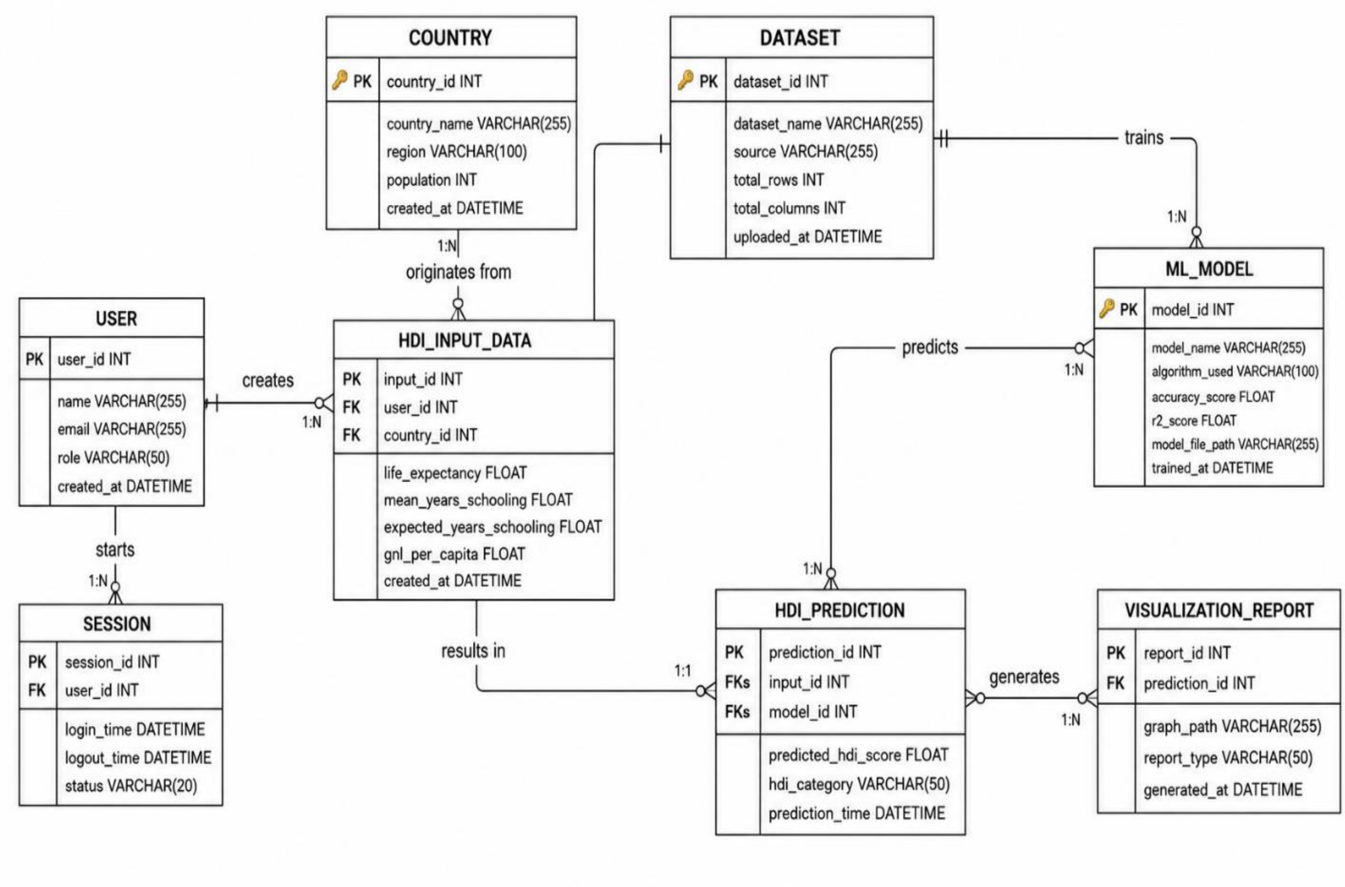

Figure 1.2: Entity Relationship Diagram

Description:

The ER Diagram shows the relationship between User Input, Machine Learning Model, Dataset, and Prediction Output. he ER diagram represents the core entities involved in the Loan Approval Prediction System and illustrates how they interact with each other. It provides a clear structure for organizing applicant information, loan requests, prediction results, and machine learning model outputs within the database.

Entities Involved

The ER diagram features eight primary entities:

User

Country

HDI Input Data

ML Model

HDI Prediction

Dataset
Visualization Report
Session

Primary Keys

Each entity is uniquely identified by its primary key:

User: user_id
Country: country_id
HDI Input Data: input_id
ML Model: model_id
HDI Prediction: prediction_id
Dataset: dataset_id
Visualization Report: report_id
Session: session_id

Relationships

User to HDI Input Data: One user can submit multiple HDI prediction inputs (1 to Many).

Country to HDI Input Data: One country can have multiple HDI input records for prediction analysis (1 to Many).

HDI Input Data to HDI Prediction: A single input record generates one HDI prediction result (1 to 1).

ML Model to HDI Prediction: One machine learning model can generate multiple HDI predictions (1 to Many).

HDI Prediction to Visualization Report: One prediction can generate multiple reports and visualizations (1 to Many).

Dataset to ML Model: One dataset can be used to train multiple machine learning models (1 to Many).

User to Session: One user can have multiple application sessions (1 to Many).

User Attributes

Attributes include:

user_id (PK)
name
email
role
created_at

13. FLOW CHART

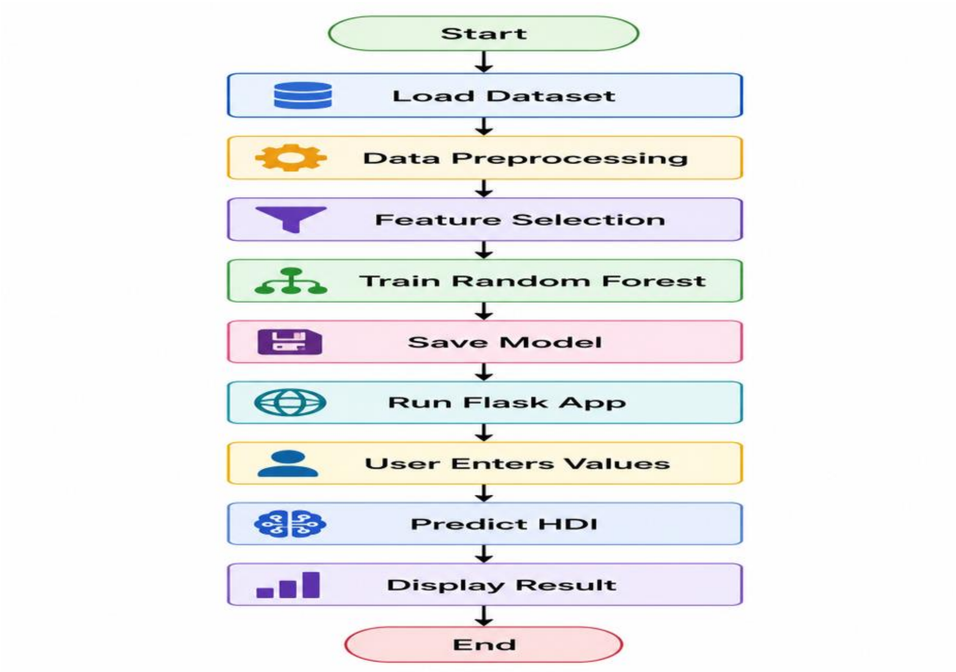

Figure 1.3: Project Flowchart

Description: The project is developed through a structured workflow consisting of multiple epics, covering data collection, analysis, preprocessing, model development, and deployment. Each epic focuses on a specific stage of the machine learning lifecycle to ensure systematic and efficient project execution.

Epic 1: Environment Setup and Package Installation

Story 1: Download and install all required Python packages, machine learning libraries, and Flask dependencies needed for the project.

Story 2: Create and organize the project folder structure, including directories for datasets, models, templates, static files, and source code.

Epic 2: Importing Required Libraries

Story 1: Import all necessary libraries and modules such as NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, Pickle, and Flask to support data analysis, model development, and web application creation.

Epic 3: Dataset Download and Understanding

Story 1: Download the dataset from Kaggle and load it into the development environment.

Story 2: Explore and understand the dataset structure, features, target variable, and data types.

Story 3: Perform data visualization to identify patterns, trends, distributions, and relationships within the dataset.

Epic 4: Data Preprocessing and Label Encoding

Story 1: Select dependent and independent variables required for model training.

Story 2: Check for missing values and handle null entries appropriately to improve data quality.
Story 3: Perform label encoding to convert categorical text values into numerical representations suitable for machine learning algorithms.
Story 4: Prepare the cleaned and transformed dataset for model development.
Epic 5: Dividing the Dataset into Train and Test Data
Story 1: Split the processed dataset into training and testing sets to enable model learning and performance evaluation on unseen data.
Epic 6: Fitting the Model
Story 1: Train the Linear Regression model using the prepared training dataset.
Story 2: Generate predictions using the trained model and analyze the prediction results.
Story 3: Evaluate model performance using appropriate regression metrics and visualizations.
Epic 7: Saving the Model
Story 1: Save the trained Linear Regression model using serialization techniques such as Pickle so it can be reused without retraining.
Story 2: Store the model file for deployment and future prediction purposes.
Epic 8: Building the Flask Web Application
Story 1: Develop the Flask backend to handle user requests, process inputs, load the trained model, and generate predictions.
Story 2: Create HTML templates and integrate them with Flask to provide a user-friendly interface.
Story 3: Run, test, and validate the complete web application to ensure accurate predictions and smooth functionality.

The flowchart explains the complete workflow starting from user input to prediction generation.

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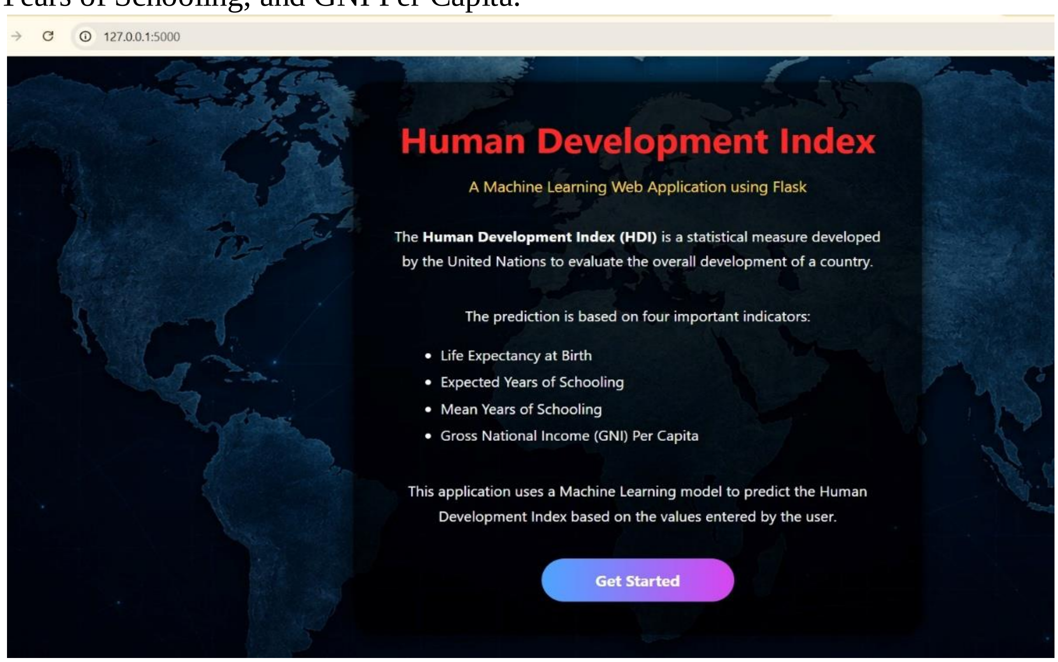
14. IMPLEMENTATION

Step 1

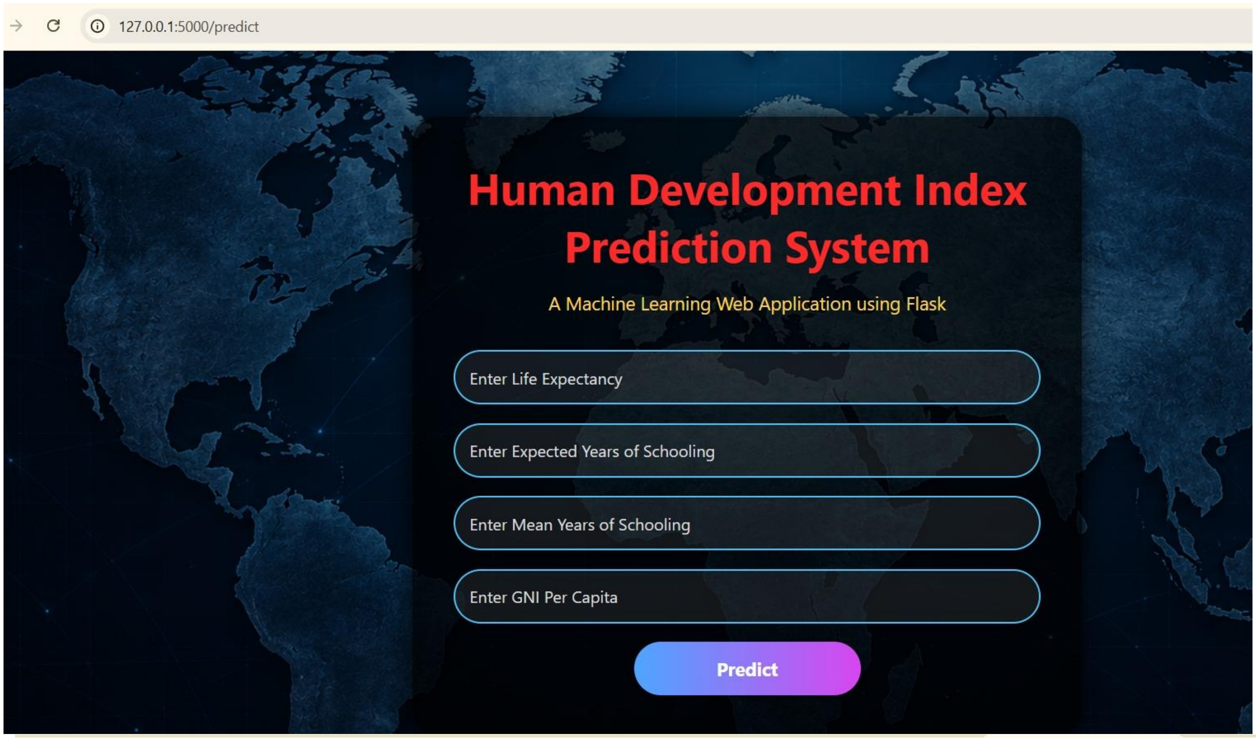
Import required libraries.
The project was developed using a set of powerful Python-based tools and libraries for data processing, machine learning, visualization, and deployment. These technologies provide the foundation for efficient model development, analysis, and application deployment.
NumPy: Core library for numerical computing in Python, providing support for large, multi-dimensional arrays and mathematical functions. https://numpy.org/doc/stable/
Pandas: Data manipulation and analysis library offering data structures and operations for manipulating numerical tables and time series. https://pandas.pydata.org/docs/
Scikit-learn: Machine learning library featuring classification, regression, and clustering algorithms like support vector machines and random forests. https://scikit-learn.org/stable/
Matplotlib: Comprehensive library for creating static, animated, and interactive visualizations in Python. https://matplotlib.org/stable/

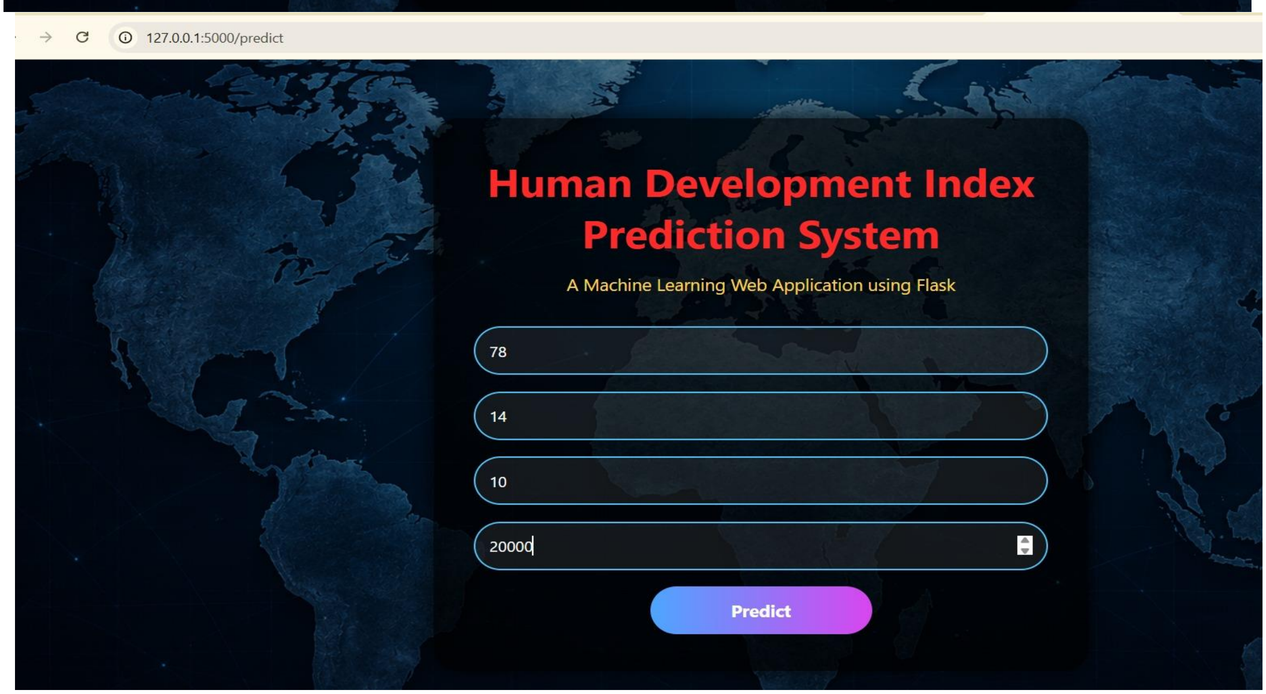
Seaborn: Data visualization library based on matplotlib that provides a high-level interface for drawing attractive statistical graphics. https://seaborn.pydata.org/
Flask: A lightweight WSGI web application framework used to build web APIs and deploy machine learning models. https://flask.palletsprojects.com/

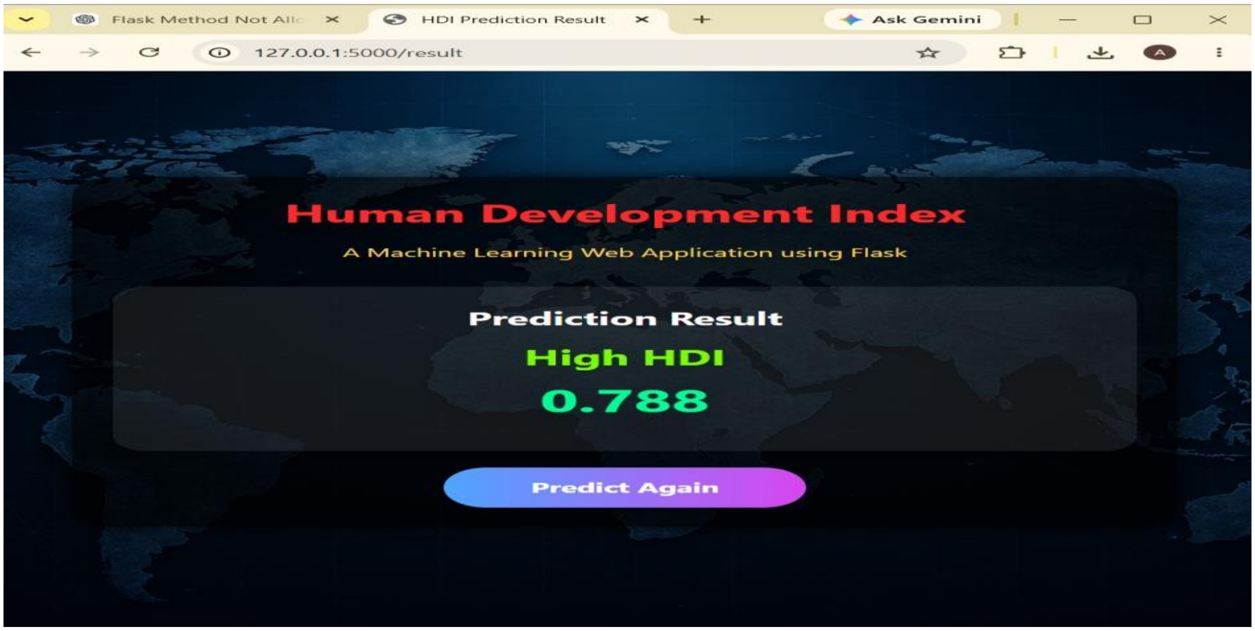
Step 2
Load dataset.
Step 3
Clean dataset.
Step 4
Train Random Forest model.
Step 5
Save model using Pickle.
Step 6
Develop Flask application.
Step 7
Run application.
15. RESULTS
The model successfully predicts Human Development Index values based on user inputs.
Prediction accuracy achieved:
99.5%
The web application provides instant prediction results.
Home Page
The Home Page allows users to enter Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, and GNI Per Capita.



Prediction Page



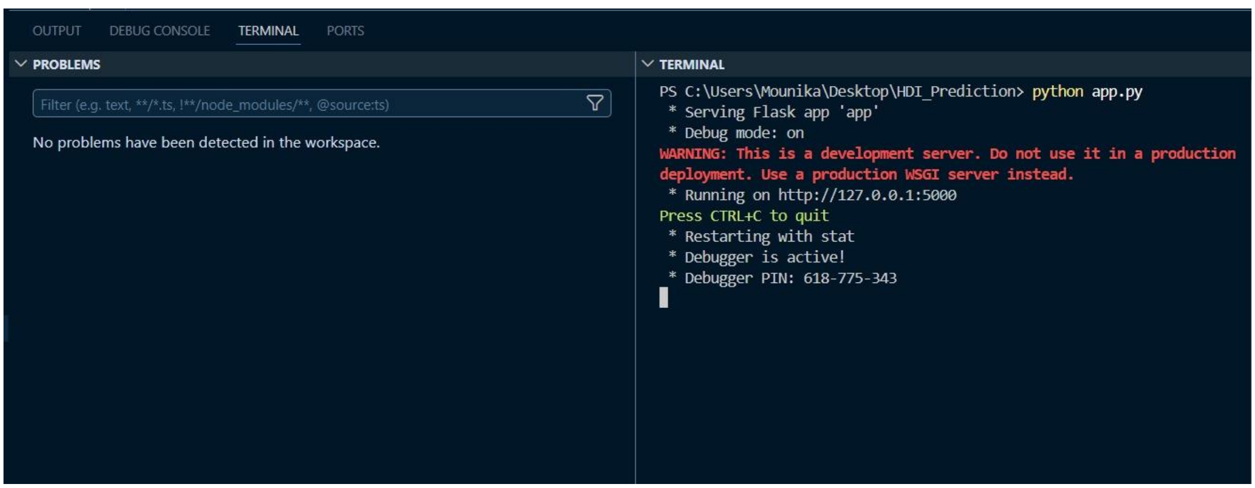




Description:

Displays the predicted HDI score generated by the trained Random Forest Regression model.

VS Code Screenshot



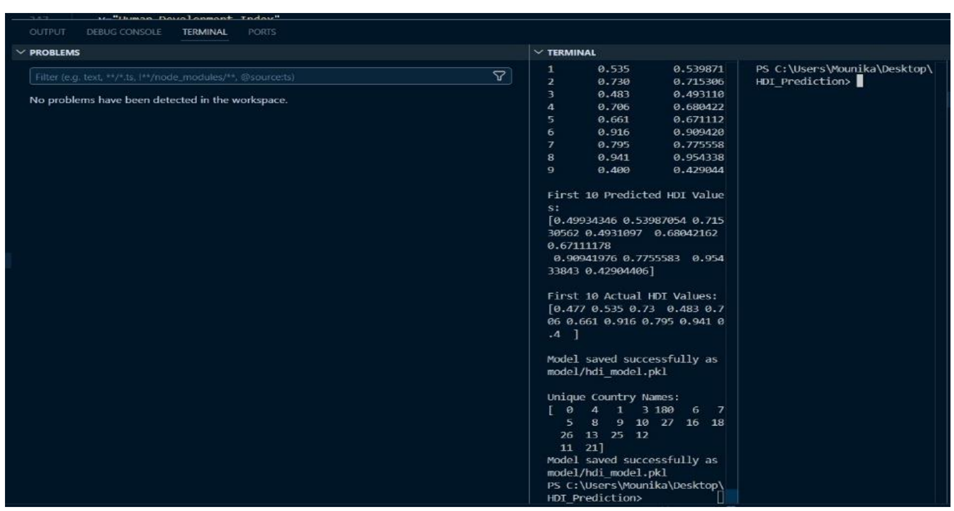
Description:

Shows the project source code in Visual Studio Code.

Terminal Screenshot

Description:

Displays successful execution of model training and Flask application.



Data Visualization:

Mean Years of Schooling vs HDI: Plot Mean Years of Schooling (X-axis) vs HDI Score (Y-axis) using a strip plot. Analyzes the education-HDI relationship.

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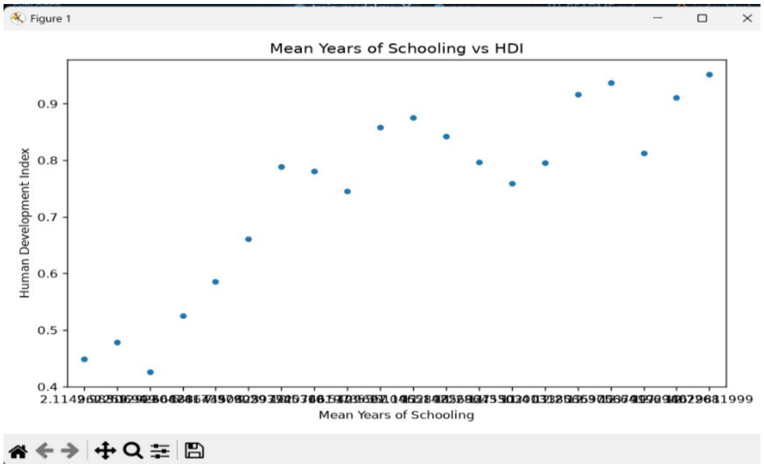
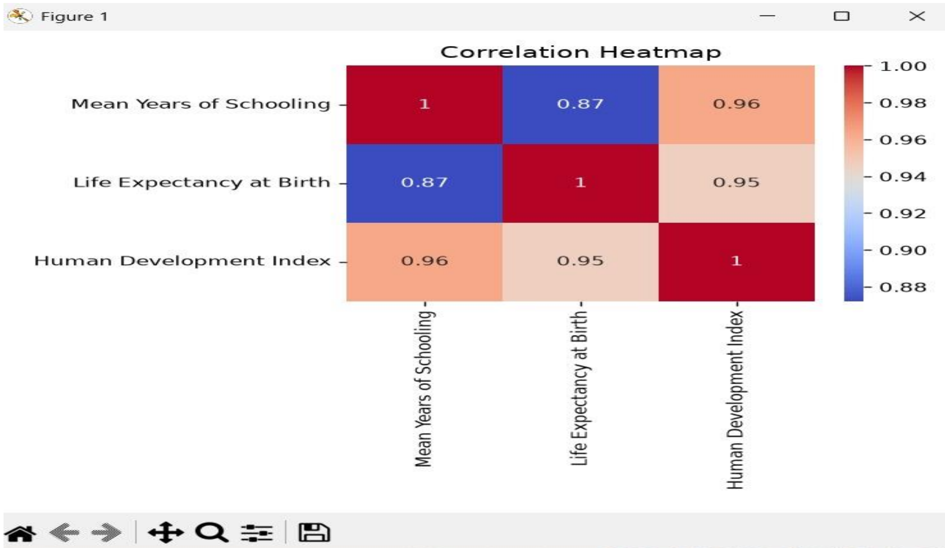
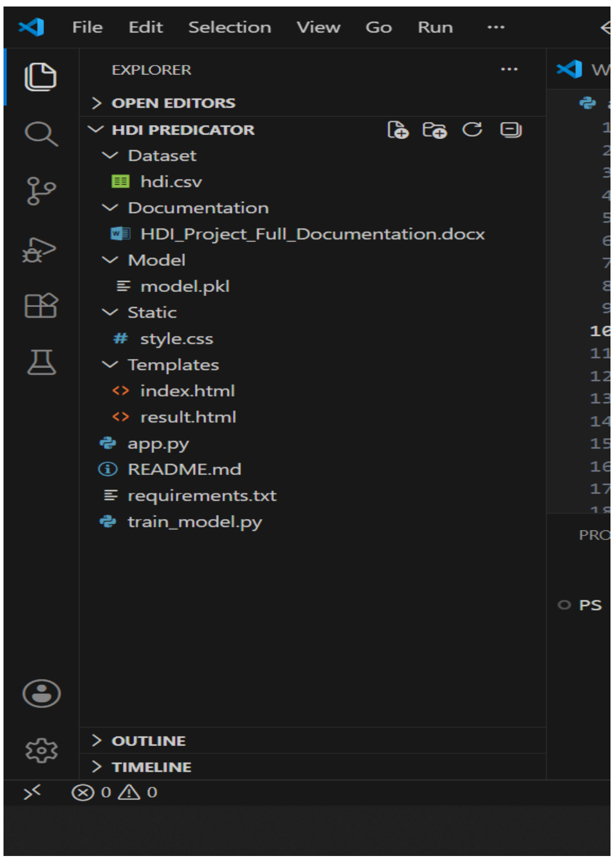


Life Expectancy vs HDI: Plot Life Expectancy (X-axis) vs HDI Score (Y-axis). Reveals how longevity correlates with human development score.

Correlation Heatmap: Build a correlation matrix heatmap using Seaborn. Each cell value represents the correlation coefficient between two features. Use this to select the most relevant input columns for model training higher correlation with HDI indicates a stronger predictor.



Folder Structure



Description:
Shows the complete project directory structure.

16. ADVANTAGES

- High Prediction Accuracy
- User Friendly
- Fast Prediction
- Easy Deployment
- Real-time Prediction
- Low Computational Cost

17. FUTURE SCOPE

The project can be extended by:

- Adding country-wise prediction.
- Deploying on cloud platforms.
- Using Deep Learning models.
- Interactive dashboard visualization.
- Real-time global data integration.

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18. CONCLUSION

The Human Development Index Prediction System successfully predicts HDI values using Machine Learning techniques. The Random Forest Regression model achieved high prediction accuracy and provides quick results through a Flask web application. The project demonstrates how Artificial Intelligence can assist policymakers, researchers, and students in understanding and evaluating human development indicators efficiently.

A Comprehensive Measure of Well-Being provides a holistic view of quality of life by evaluating multiple dimensions that influence an individual's overall welfare, rather than relying solely on traditional economic indicators such as income or GDP. It encompasses a wide range of factors, including physical and mental health, educational attainment, financial stability, employment opportunities, social relationships, environmental quality, personal safety, and overall life satisfaction.

By integrating these diverse aspects, a comprehensive well-being framework offers a more accurate and meaningful assessment of how individuals and communities are truly thriving. It helps uncover hidden challenges that may not be reflected through economic measures alone, while also highlighting strengths and opportunities for growth.

This multidimensional approach enables policymakers, researchers, healthcare professionals, and organizations to make informed decisions, design effective interventions, and allocate resources more efficiently. It also supports the development of strategies aimed at improving public health, reducing inequalities, enhancing social cohesion, and promoting sustainable development.

Ultimately, measuring well-being in a comprehensive manner contributes to building healthier, happier, and more resilient societies. By focusing on the factors that genuinely impact people's lives, it encourages balanced progress and helps create inclusive communities where individuals have the opportunity to achieve their full potential and enjoy a higher quality of life.

19. REFERENCES

1. UNDP Human Development Reports
2. Scikit-Learn Documentation
3. Pandas Documentation
4. Flask Documentation
5. NumPy Documentation